

MATH70013 Advanced Simulation Methods

Coursework 1 Solution

1. Parameter inference for the linear Gaussian model

I generated one dataset with $T = 100$, true parameters $(\rho^*, \tau^*, \sigma) = (0.8, 1, 0.5)$, and stored the latent trajectory $x_{0:T}^*$ in `generated_data_assignment1_26.npz`.

1.1 MCMC method and design

I used a random-walk Metropolis–Hastings algorithm on $\theta = (\rho, \tau)$ with 20,000 iterations.

$$\begin{aligned}\rho' &= \rho + \epsilon_\rho, & \epsilon_\rho &\sim \mathcal{N}(0, s_\rho^2), & s_\rho &= 0.12, \\ \tau' &= \tau + \epsilon_\tau, & \epsilon_\tau &\sim \mathcal{N}(0, s_\tau^2), & s_\tau &= 0.18.\end{aligned}$$

If $\rho' \notin [-1, 1]$ or $\tau' \notin (0, 2]$, the proposal is rejected. Priors are:

$$\begin{aligned}\rho &\sim \mathcal{U}[-1, 1], \\ \tau &\sim \mathcal{N}_{(0,2]}(1, 1).\end{aligned}$$

The likelihood $p(y_{0:T} \mid \rho, \tau)$ is computed exactly with the Kalman filter (integrating out $x_{0:T}$), and MH uses the standard acceptance ratio on log posterior values.

Acceptance rate was 0.3639. I discarded the first 5,000 draws as burn-in.

1.2 Marginal approximations

Figure 1 shows the posterior marginals for ρ and τ .

- Posterior mean: $\mathbb{E}[\rho \mid y] \approx 0.738$, $\mathbb{E}[\tau \mid y] \approx 1.208$.
- 95% credible intervals: $\rho \in [0.589, 0.885]$, $\tau \in [1.014, 1.438]$.
- Relative to true values, the posterior centers are shifted by -0.062 for ρ (about 7.7% of 0.8) and $+0.208$ for τ (about 20.8% of 1).

Interpretation from this specific realization: the stored latent trajectory has empirical innovation scale

$$\text{sd}(x_n^* - 0.8x_{n-1}^*) \approx 1.182,$$

which is above $\tau^* = 1$, so the posterior moving toward larger τ is consistent with the realized path. The posterior is also much tighter than the priors: for ρ , prior sd is $\sqrt{1/3} = 0.577$ (uniform prior) versus posterior sd 0.076 (a $7.6\times$ contraction); for τ , prior sd is 0.540 (truncated normal prior) versus posterior sd 0.108 (a $5.0\times$ contraction). In precision terms (inverse variance), this is about $57\times$ gain for ρ and $25\times$ for τ , which is a direct empirical sign of higher observed Fisher information in the ρ direction for this dataset.

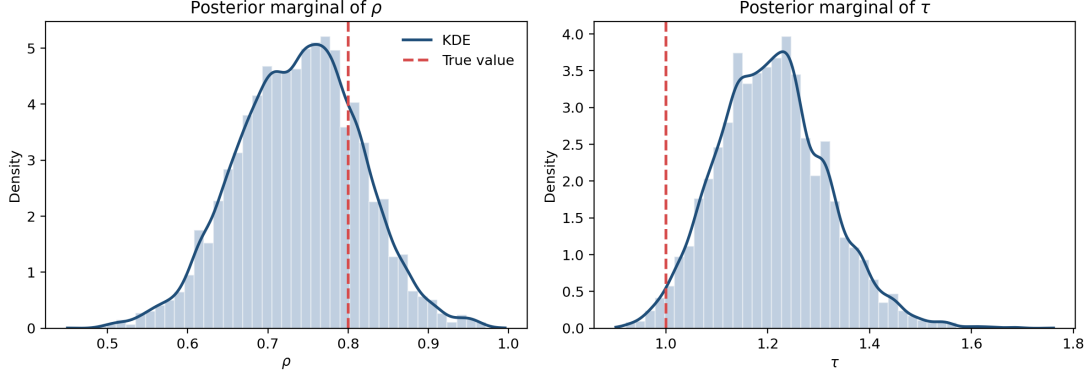


Figure 1: Posterior marginal approximations for ρ and τ .

1.3 Diagnostics and quality of approximation

I report two diagnostics:

- Trace plots for ρ and τ .
- Autocorrelation function (ACF) for post-burn-in samples.

These are shown in Figure 2. The chains appear stationary after burn-in with regular exploration of the high-posterior region. Quantitatively, ACF is 0.784/0.764 at lag 1 (ρ/τ), drops to 0.122/0.074 by lag 10, and is near zero by lag 20 (0.028/−0.024), which matches the visual decay in the plots. Effective sample sizes (post-burn-in) are:

$$\text{ESS}(\rho) \approx 1577, \quad \text{ESS}(\tau) \approx 2024$$

out of 15,000 retained samples (Monte Carlo efficiency about 10.5% and 13.5%, respectively). Equivalently, integrated autocorrelation times are about 9.5 for ρ and 7.4 for τ . For a 2D posterior, this is adequate mixing and supports reliable marginal estimation.

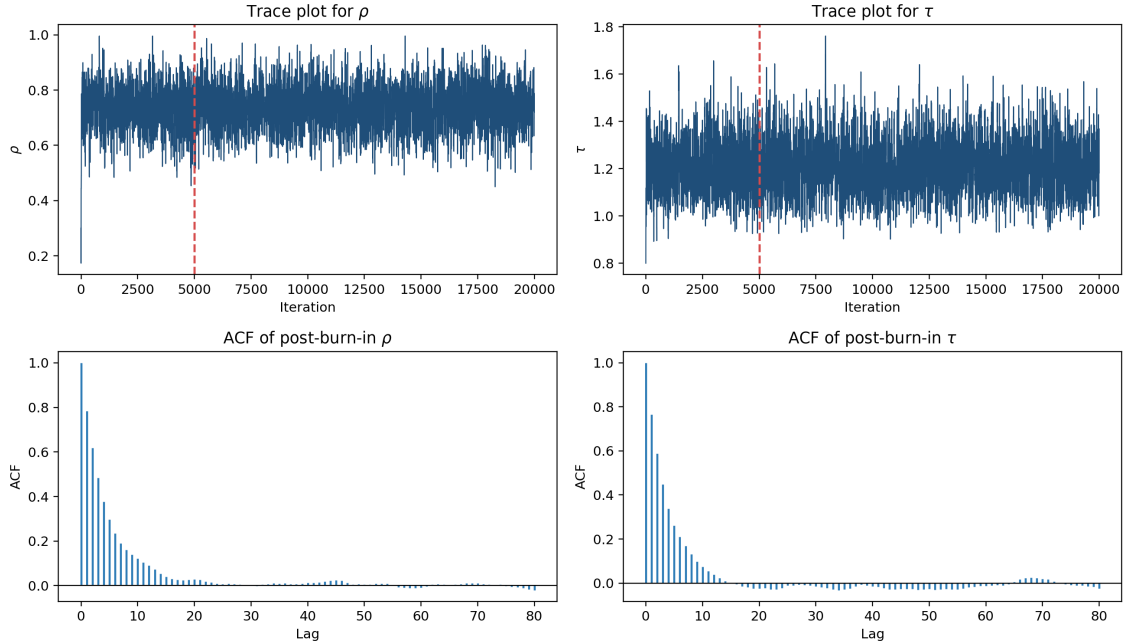


Figure 2: Trace plots and ACF diagnostics for MH output. Red dashed line marks burn-in end.

1.4 Effect of changing σ (no additional simulations)

σ controls observation noise in $Y_n = X_n + \sigma W_n$.

- Smaller σ : each observation carries more information about the latent state; in information terms, Gaussian observation Fisher information scales as $1/\sigma^2$, so halving σ multiplies this contribution by about 4. Starting from the current posterior widths (CI widths 0.296 for ρ and 0.424 for τ), we would therefore expect visibly narrower marginals and usually lower MH acceptance if proposal scales were not retuned.
- Larger σ : observation information decreases (e.g., doubling σ reduces this component by about $4\times$), so posteriors become broader and more prior-driven. The current run already shows τ less tightly identified than ρ ; increasing σ would amplify that gap and strengthen the ρ - τ confounding between persistence and process-noise amplitude.

2. Testing particle filters with a linear Gaussian model

2.1 Choice of importance density

I chose the locally optimal proposal

$$q(x_n \mid y_n, x_{n-1}) = p(x_n \mid y_n, x_{n-1}).$$

For this model,

$$q(x_n \mid y_n, x_{n-1}) = \mathcal{N}(\mu_n^q, V_n^q), \quad V_n^q = \frac{\tau^2 \sigma^2}{\tau^2 + \sigma^2}, \quad \mu_n^q = \frac{\sigma^2 \rho x_{n-1} + \tau^2 y_n}{\tau^2 + \sigma^2}.$$

Reason: this minimizes variance of incremental importance weights conditionally on (x_{n-1}, y_n) , so it is a principled and efficient choice for both SIS and SIR.

2.2 Filtering density approximations at $n = 50$ and $n = 100$

Figures 3 and 4 compare SIS and SIR for $N = 50$ and $N = 500$. Each panel overlays particle approximation and Kalman filtering density (exact benchmark). At $n = 50$, all methods except SIS- $N = 50$ are reasonably close, but the plot already shows the key trade-off: SIR- $N = 500$ is slightly flatter around the mode than SIS- $N = 500$, consistent with additional resampling variance. At $n = 100$, long-horizon behavior dominates: SIS- $N = 50$ is clearly collapsed (overly narrow, wrong shape), while SIR- $N = 500$ is almost indistinguishable from the Kalman benchmark.

Using the same KDE curves as in the figures, integrated squared error (ISE) against the Kalman density is:

ISE at $n = 50$: 0.0598 (SIS-50), 0.00571 (SIS-500), 0.00249 (SIR-50), 0.0123 (SIR-500),

ISE at $n = 100$: 0.5638 (SIS-50), 0.0130 (SIS-500), 0.0283 (SIR-50), 0.000524 (SIR-500).

So SIR- $N = 500$ is temporarily less accurate at $n = 50$ (resampling noise cost), but by $n = 100$ it is about $24.8\times$ better than SIS- $N = 500$ on ISE, reflecting superior long-run stability.

Filtering density approximations at $n=50$

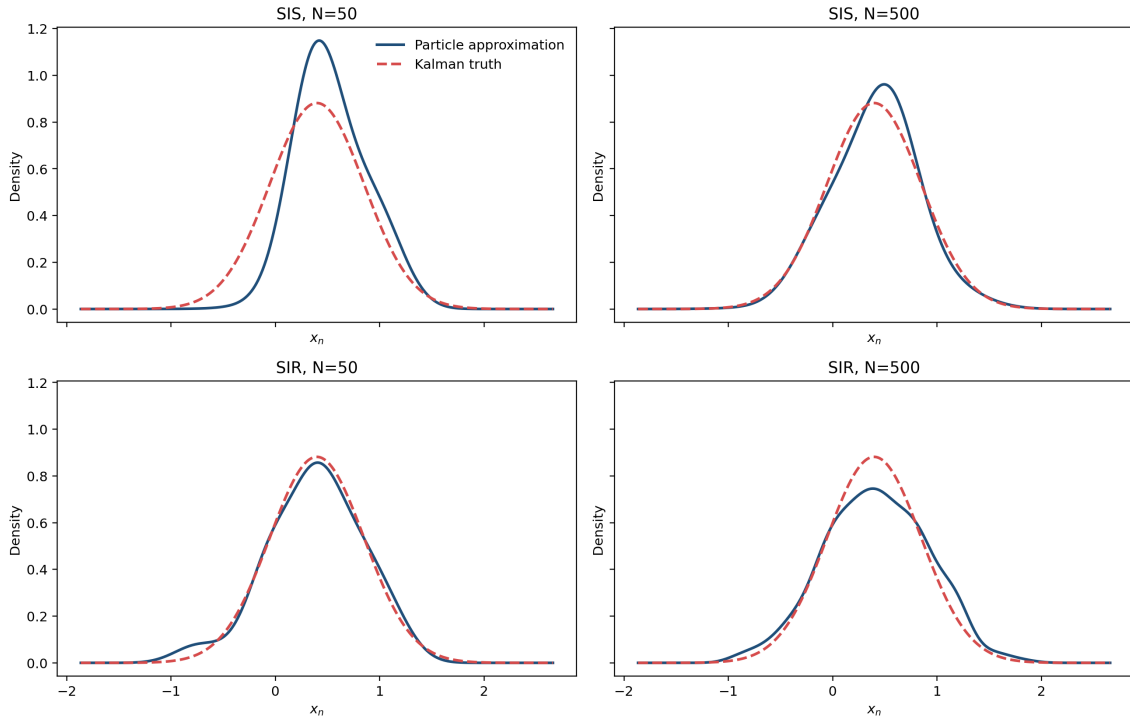


Figure 3: Approximations of $p(x_{50} | y_{0:50})$ for SIS/SIR and $N \in \{50, 500\}$.

Filtering density approximations at $n=100$

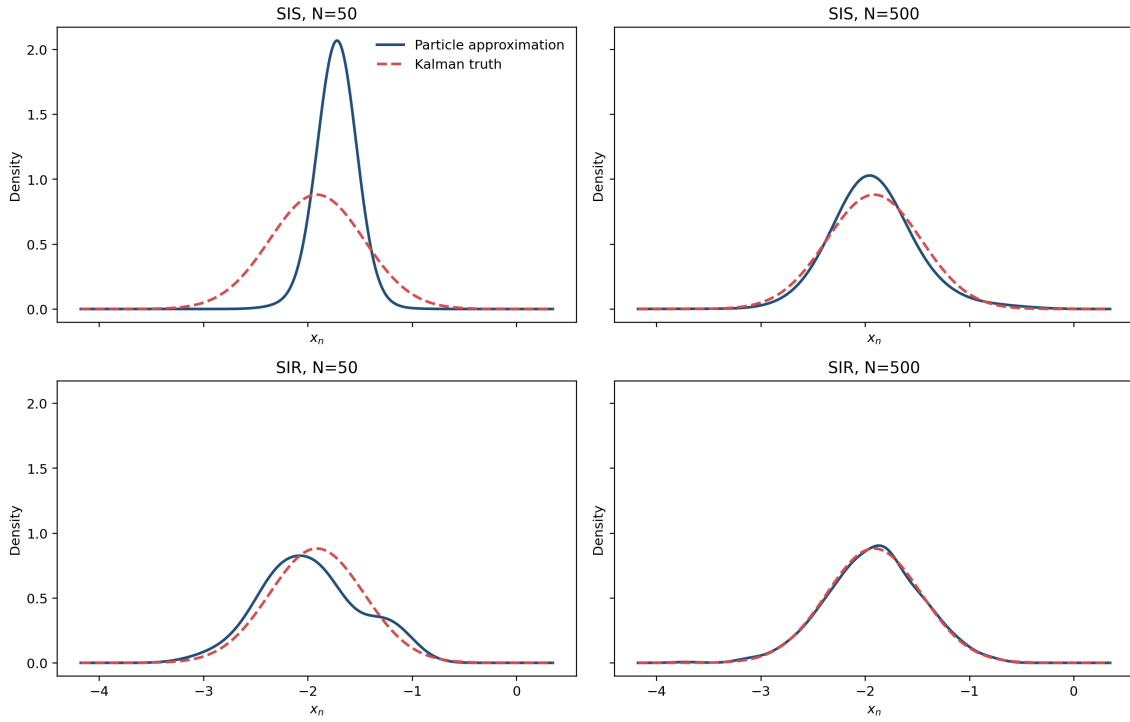


Figure 4: Approximations of $p(x_{100} | y_{0:100})$ for SIS/SIR and $N \in \{50, 500\}$.

2.3 Estimating $P(|X_n| \leq 0.1 \mid y_{0:n})$ for $n \leq T$

For particles $\{x_n^{(i)}, w_n^{(i)}\}_{i=1}^N$, I used

$$\hat{P}_n = \sum_{i=1}^N w_n^{(i)} \mathbf{1}\{|x_n^{(i)}| \leq 0.1\}.$$

The exact benchmark is from Kalman filtering: $X_n \mid y_{0:n} \sim \mathcal{N}(m_n, C_n)$, so

$$P(|X_n| \leq 0.1 \mid y_{0:n}) = \Phi\left(\frac{0.1 - m_n}{\sqrt{C_n}}\right) - \Phi\left(\frac{-0.1 - m_n}{\sqrt{C_n}}\right).$$

Figure 5 shows all methods against the exact curve.

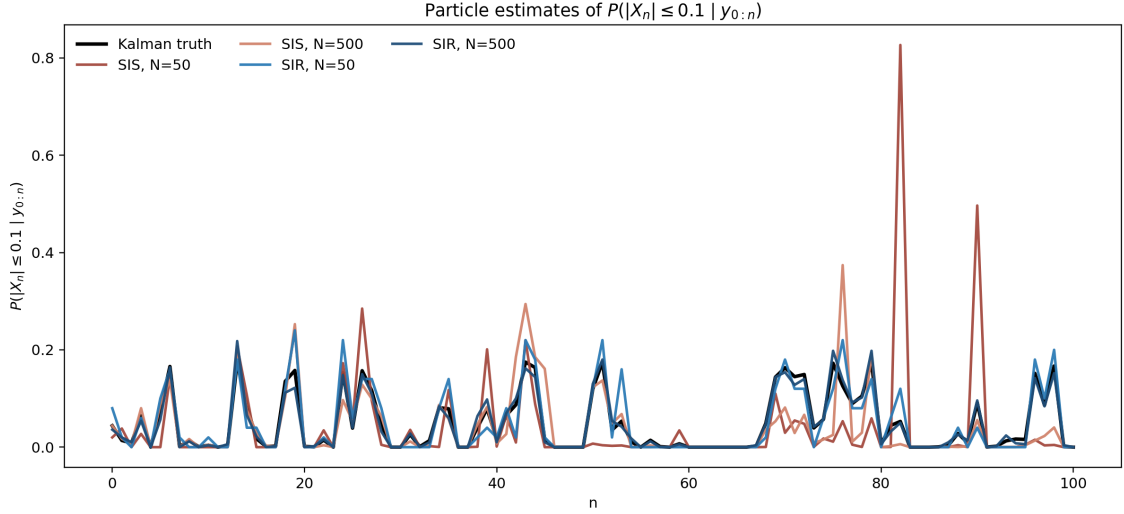


Figure 5: Estimated and exact $P(|X_n| \leq 0.1 \mid y_{0:n})$ over time.

A summary of error and degeneracy indicators:

Method	RMSE of $\hat{P}_n$ vs exact	RMSE gain vs SIS (same N)	Final ESS / N
SIS, $N = 50$	0.1018	1.00×	1.289/50 = 2.58%
SIS, $N = 500$	0.0494	1.00×	2.912/500 = 0.58%
SIR, $N = 50$	0.0274	3.71×	50/50 = 100%
SIR, $N = 500$	0.0102	4.83×	500/500 = 100%

The probability-path figure confirms this numerically. Example: SIS- $N = 50$ has a large spurious spike $\hat{P}_{82} = 0.826$ while the exact value is 0.053 (absolute error 0.774). For SIR- $N = 500$, the maximum absolute error over all n is only 0.0436, about 17.7× smaller than that SIS spike. Also, SIR- $N = 50$ (10× fewer particles) still outperforms SIS- $N = 500$ on RMSE by a factor 1.80×

2.4 Strengths of SIR over SIS observed here

The main strengths of SIR are clearly visible and align with the course theory on weight degeneracy:

- SIR controls weight degeneracy: ESS/ N remains at 100% after each resampling step, while SIS drops to 2.58% ($N = 50$) and 0.58% ($N = 500$) at the final time.

- In SIS, final weights are highly concentrated (e.g., for $N = 500$, the largest particle weight is 0.562 and the top 5 weights already sum to 0.817), which is exactly the path-degeneracy mechanism discussed in lectures.
- This translates into better estimation accuracy: for the probability functional, SIR improves RMSE by $3.71\times$ ($N = 50$) and $4.83\times$ ($N = 500$) versus SIS.

Trade-off (also visible in the figures): resampling introduces variance and can slightly distort an individual-time density (e.g., SIR- $N = 500$ at $n = 50$ is flatter than Kalman), but it prevents catastrophic weight collapse. Over longer horizons (e.g., $n = 100$ and the full probability trajectory), that trade-off is strongly favorable to SIR in this experiment.